1. What is SQL? Explain briefly.

SQL (Structured Query Language) is a standard programming language used to manage and manipulate relational databases.

It allows users to create, retrieve, update, and delete data stored in database tables.

SQL is also used to define database structures, control access permissions, and perform operations such as filtering, sorting, grouping, and joining data.

It is widely used in relational database management systems like MySQL, Oracle, SQL Server, and PostgreSQL.

2. What is the difference between DDL, DML, and DCL?

DDL (Data Definition Language):
DDL commands are used to define and modify the structure of database objects such as tables, schemas, and indexes. These commands affect the structure of the database.
Examples: CREATE, ALTER, DROP, TRUNCATE.

DML (Data Manipulation Language):
DML commands are used to manipulate the data stored inside the database tables. These commands help insert, update, retrieve, or delete records from tables.
Examples: INSERT, UPDATE, DELETE, SELECT.

DCL (Data Control Language):
DCL commands are used to control access permissions and security in a database. They determine which users can access or modify database objects.
Examples: GRANT, REVOKE.

3. What is Data?
Data refers to raw facts and figures that can be processed to generate meaningful information. In databases, data is stored in tables in the form of rows and columns. Examples include names, numbers, addresses, and dates.

4. What is a Database?
A database is an organized collection of structured data that is stored electronically and can be easily accessed, managed, and updated. Databases help store large amounts of data efficiently and allow users to retrieve information quickly.

5. What is an Index?
An index is a database object used to improve the speed of data retrieval operations on a table. It works similarly to an index in a book, allowing the database to locate specific records quickly without scanning the entire table.

6. What are Joins?
Joins are SQL operations used to combine rows from two or more tables based on a related column between them. Joins allow data from multiple tables to be retrieved in a single query.

Common types of joins include:

- Inner Join: Returns records that have matching values in both tables.
- Left Join: Returns all records from the left table and matched records from the right table.
- Right Join: Returns all records from the right table and matched records from the left table.
- Full Join: Returns all records when there is a match in either table.
- Self-join: Returns records by joining a table with itself based on a related column within the same table.
- Cross-join: Returns all possible combinations of records from both tables (Cartesian product).

7. What is a Datatype? Explain different datatypes in SQL.
A datatype defines the type of value that can be stored in a table column. It ensures that the data entered into the column follows a specific format.

Common SQL datatypes include:

-Numeric Datatypes:
Used to store numbers.
Examples:
INT – Stores integer values
FLOAT – Stores decimal values
DECIMAL – Stores fixed decimal numbers

-Character Datatypes
Used to store text values.
Examples:
CHAR – Fixed-length character string
VARCHAR – Variable-length character string
TEXT – Large text data

-Date and Time Datatypes
Used to store date and time values.
Examples:
DATE – Stores date values
TIME – Stores time values
DATETIME – Stores both date and time

-Boolean Datatype
Used to store true or false values.

8. What is an Operator? Explain different operators available in SQL.
Operators are symbols or keywords used to perform operations on data in SQL queries.

Types of SQL operators include:

-Arithmetic Operators
Used for mathematical calculations.
Examples: +, -, *, /

-Comparison Operators
Used to compare two values.
Examples: =, >, <, >=, <=, <>

-Logical Operators
Used to combine multiple conditions.
Examples: AND, OR, NOT

-Special Operators
Examples:
BETWEEN – Checks if a value is within a range
IN – Checks if a value matches any value in a list
LIKE – Searches for a specific pattern in a column

9. What is a View?
A view is a virtual table created from the result of a SQL query. It does not store data physically but displays data from one or more tables. Views are used to simplify complex queries and provide data security.

10. What are different commands available in SQL?
SQL commands are categorized into different types:

DDL Commands
CREATE, ALTER, DROP, TRUNCATE

DML Commands
INSERT, UPDATE, DELETE, SELECT

DCL Commands
GRANT, REVOKE

TCL (Transaction Control Language) Commands
COMMIT, ROLLBACK, SAVEPOINT

11. What is DBMS?
DBMS (Database Management System) is software used to create, manage, and control databases. It provides an interface for users to store, retrieve, and manipulate data efficiently while ensuring data security and integrity.

12. What is ER Model?
The ER (Entity-Relationship) model is a conceptual model used to design databases. It visually represents the structure of a database using entities, attributes, and relationships between entities. ER diagrams help in understanding how data is organized before creating the actual database.

13. How SQL Works? Explain the procedure.
The working of SQL generally follows these steps:

1. The user writes an SQL query.
2. The database parser checks the query for syntax errors.
3. The query optimizer determines the most efficient way to execute the query.
4. The query execution engine processes the query.
5. The DBMS retrieves or modifies the required data from the database.
6. The results are returned to the user.

14. What is a Table?
A table is the basic structure used to store data in a relational database. It consists of rows and columns. Each row represents a record, and each column represents an attribute of the data.

15. What is the difference between DROP, TRUNCATE, and DELETE?

DELETE
Removes selected rows from a table based on a condition. The table structure remains unchanged and the operation can be rolled back.

TRUNCATE
Removes all rows from a table quickly. It does not allow conditions and cannot usually be rolled back.

DROP
Completely removes the table from the database, including its structure and data.

16. What is an Entity?
An entity is an object or concept in the real world that can be identified and stored in a database. For example, a student, employee, or product can be considered an entity.

17. What is an Attribute?
An attribute is a property or characteristic of an entity. For example, for a student entity, attributes may include student_id, name, age, and department.

18. What are Strong and Weak Entities?

Strong Entity
A strong entity can exist independently and has its own primary key. For example, a Student table with Student_ID as the primary key.

Weak Entity
A weak entity depends on another entity for its existence and does not have a complete primary key on its own. It usually uses a foreign key along with its own partial key.

19. What are different types of relationships available?

One-to-One (1:1)
One record in a table is related to one record in another table.

One-to-Many (1:M)
One record in a table can be related to multiple records in another table.

Many-to-Many (M:N)
Multiple records in one table can relate to multiple records in another table.

20. How to Create a Database?

Example SQL command:

CREATE DATABASE college_db;

This command creates a new database named college_db.

21. How to Drop a Database?

Example SQL command:

DROP DATABASE college_db;

This command permanently removes the database and all its data from the system.